

DATA ANALYTICS INTERNSHIP

ASSIGNMENT 2

Python, NumPy, Pandas, Data Analysis & Visualization

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Course: Data Analytics Internship
Assignment No.: 2

Task 1 – Python Basics & NumPy Operations

1.1 Python Basics

```
i = 25
f = 3.14
s = "Data Analytics"
b = True
lst = [10, 20, 30]
tup = (1, 2, 3)
dct = {"name": "Aarav", "age": 28}

for value in [i, f, s, b, lst, tup, dct]:
    print(value, type(value))

a, b_num = 10, 3
print("Add:", a+b_num); print("Sub:", a-b_num); print("Mul:", a*b_num)
print("Div:", a/b_num); print("Mod:", a%b_num); print("Power:", a**b_num); print("Floor div:",
a//b_num)
print("Equal:", a==b_num); print("Greater:", a>b_num); print("Less:", a<b_num)
print("AND:", a>5 and b_num<5); print("OR:", a<5 or b_num<5)
```

1.2 NumPy Results

Shape: (12,) | Size: 12 | Dtype: int64
Sum: 2311 | Mean: 192.58 | Max: 275 | Min: 125 | Std: 47.3998
Indices > 200: [7, 8, 9, 10, 11]
Month-over-month growth: [13, 4, 13, 13, 12, 15, 15, 15, 15, 18, 17]

Reshaped 4 × 3 array:
[[125 138 142]
 [155 168 180]
 [195 210 225]
 [240 258 275]]

Task 2 – Pandas: Data Loading, Cleaning & Merging

Customers and Purchases were loaded into customers_df and purchases_df. Missing age was filled with the mean and converted to integer; missing gender was replaced with 'Unknown'. The DataFrames were merged on cust_id using an inner join.

Merged DataFrame shape: (10, 9). The required head(), info(), missing-value checks and merged head are included in the Jupyter Notebook.

Task 3 – Filtering, Grouping, Aggregation & Visualization

3.1 Filtering

The notebook contains outputs for Mumbai purchases, purchases with amount > ₹5,000, and male customers who made any purchase.

3.2 Grouping & Aggregation

Total purchase amount per city

Bangalore: 34,699

Delhi: 8,499

Mumbai: 73,000

Pune: 2,150

Average purchase amount per gender

F: 2,833.00

M: 15,692.71

Number of purchases per customer

C01: 2

C02: 2

C03: 2

C05: 2

C06: 1

C07: 1

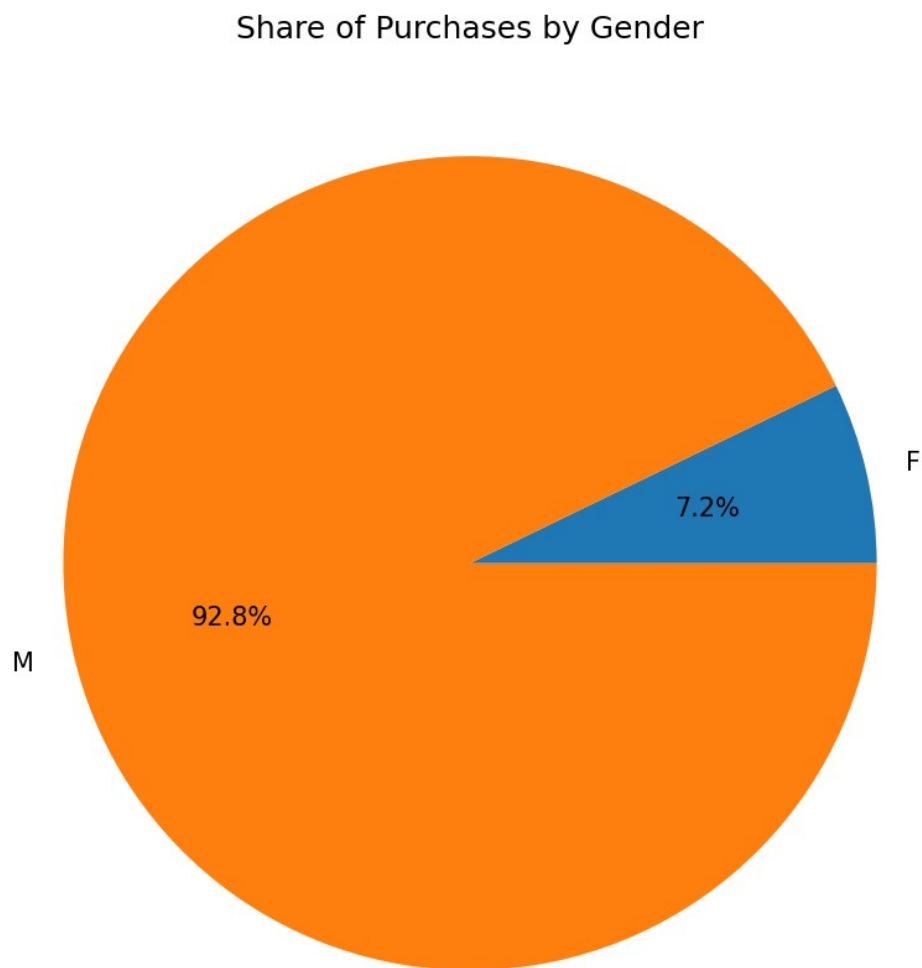
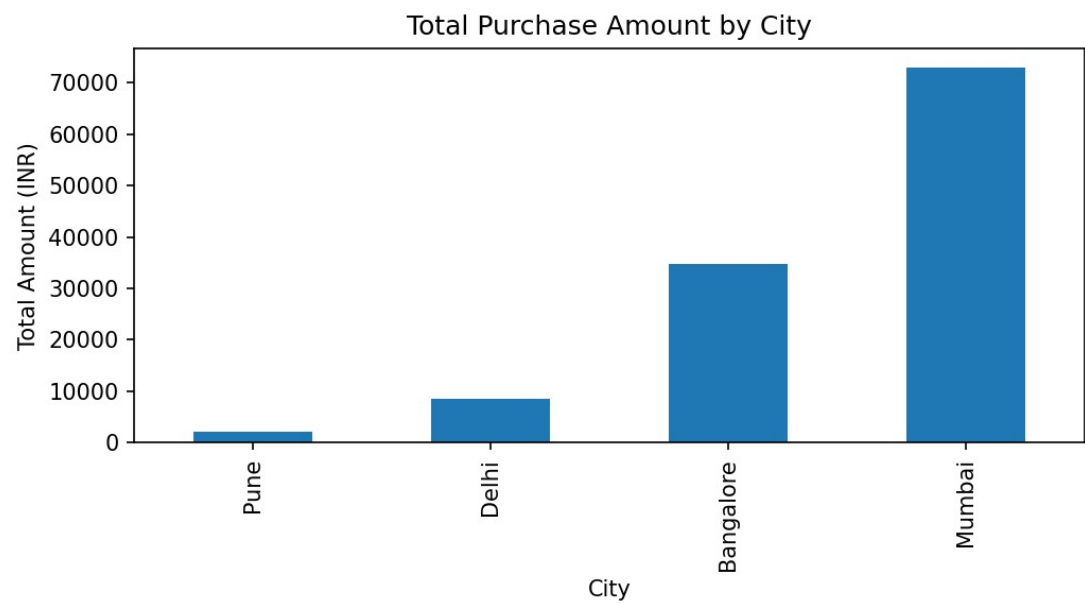
Top 3 customers by total spending

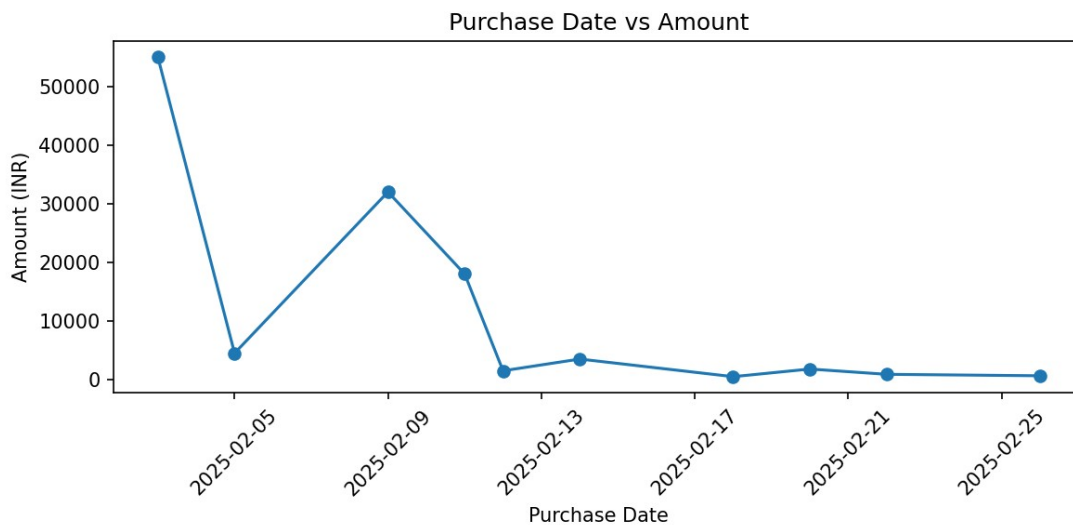
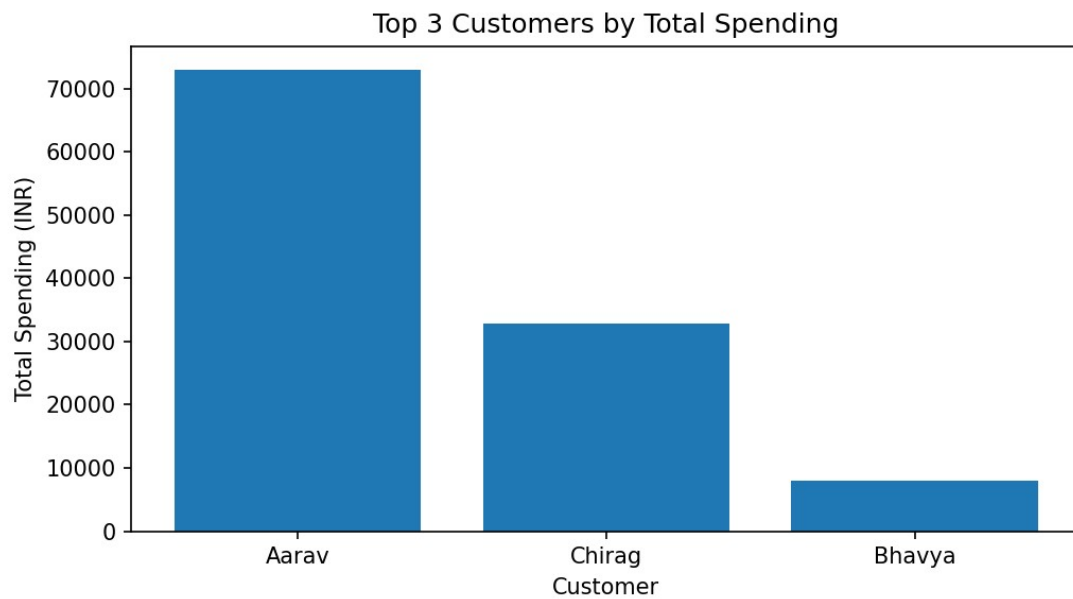
Aarav: 73,000

Chirag: 32,899

Bhavya: 8,000

3.3 Data Visualization





3.4 Data Analysis Summary

Mumbai generated the highest purchase revenue at ₹73,000, making it the strongest-performing city in the dataset. Aarav and Chirag were the most active customers by number of purchases, with two purchases each. Aarav was the top customer by total spending at ₹73,000, followed by Chirag at ₹32,899 and Bhavya at ₹8,000. The purchase pattern is concentrated in a few higher-value electronics purchases, especially the Laptop and Smartphone. Male customers recorded a higher average purchase amount than female customers in this sample. After cleaning, the missing gender value was replaced with Unknown and the missing age was filled using the mean age. Overall, the analysis shows that city, customer and product mix have a strong effect on total revenue.

Conclusion

The completed Jupyter Notebook contains the Python code, outputs, filtering/grouping analysis and all four required visualizations.